

Medvale Internal Medicine

Chapter 1J: Adult Congenital Heart Disease

Paragraph-heavy study chapter for graph-RAG indexing, with numeric anchors and original diagrams/tables

Why this matters clinically

Congenital heart disease is often taught as pediatrics, but the internal medicine clinician sees it in adults in several forms: an unrepaired lesion that remained silent for decades, a repaired childhood lesion with late complications, or a lesion discovered incidentally during workup of dyspnea, murmur, stroke, cyanosis, arrhythmia, or pulmonary hypertension. The adult version of congenital heart disease is therefore not just anatomy. It is physiology over time. A small left-to-right shunt may be clinically quiet, while a larger shunt can remodel the right heart, increase pulmonary blood flow, and eventually cause pulmonary vascular disease. Once pulmonary vascular resistance rises enough, the direction of flow may reverse, producing Eisenmenger physiology with cyanosis and major treatment implications.

This chapter focuses on the congenital lesions most relevant to an internal medicine cardiology unit: atrial septal defect, ventricular septal defect, coarctation of the aorta, patent ductus arteriosus, and tetralogy of Fallot. The uploaded supplemental section emphasizes exactly these lesions and gives high-yield anchors such as ostium secundum ASD as about 80% of ASDs, ASD symptoms often appearing around age 40, pulmonary-to-systemic flow thresholds such as Qp:Qs >1.5:1 or 2:1 for closure decisions, coarctation findings of upper-extremity hypertension with delayed femoral pulses, PDA machinery murmur, and the four components of tetralogy of Fallot.

Core principle

Most adult congenital lesions are best understood by answering four questions: Where is the abnormal connection or obstruction? Which chamber receives extra volume or pressure? Is pulmonary blood flow increased or decreased? Has pulmonary hypertension or shunt reversal developed?

Numbers to Know: Adult Congenital Heart Disease

Topic	Number anchor	Clinical meaning
Adult CHD burden in adult heart disease	About 2% in the uploaded source	Low prevalence but high-yield because findings are distinctive
ASD subtype	Ostium secundum about 80% of ASDs	Most common ASD; central interatrial septum
ASD symptom timing	Often asymptomatic until around age 40	Adults may present with exercise intolerance, dyspnea, fatigue, AFib
ASD right-heart dilation threshold	Qp:Qs >1.5:1 in source	Significant left-to-right shunt causing RA/RV volume load
ASD closure threshold	Qp:Qs >1.5:1 or 2:1, or symptomatic	Closure considered when shunt is hemodynamically significant and pulmonary vascular disease is not prohibitive
ASD pulmonary hypertension	Does not usually occur before age 20; common by age 40 in source	Late complication, not an early feature of uncomplicated ASD
PDA fetal closure	Usually closes within days after birth	Persistent ductus creates postnatal left-to-right shunt
Perinatal PDA manipulation	Indomethacin closes; prostaglandin E1 keeps open	Classic pharmacology anchor
Tetralogy of Fallot components	4 defects	VSD, RV hypertrophy, pulmonary stenosis/atresia, overriding aorta
TOF repair timing in source	Usually within first year of life	Adults often have repaired TOF and late complications
TOF post-surgery survival in source	>80% 20-year survival	Late surveillance still needed for arrhythmias and pulmonary regurgitation

Core hemodynamic vocabulary

A shunt is an abnormal pathway that allows blood to move between circulations. A left-to-right shunt sends oxygenated systemic-side blood back into the pulmonary circulation. This increases pulmonary blood flow and loads the right-sided chambers. ASD, VSD, and PDA all usually begin as left-to-right shunts after birth because left-sided pressures exceed right-sided pressures. The clinical consequences depend on the shunt size, pressure gradient, duration, and pulmonary vascular response.

The ratio $Q_p:Q_s$ compares pulmonary blood flow to systemic blood flow. A ratio of 1:1 means pulmonary and systemic flow are equal. A ratio above 1 means more blood is recirculating through the lungs than reaching systemic circulation. In many adult congenital discussions, $Q_p:Q_s$ greater than 1.5:1 marks a hemodynamically important shunt, although closure decisions also depend on symptoms, right-sided chamber enlargement, pulmonary vascular resistance, and expert congenital cardiology evaluation.

Eisenmenger physiology is a late and dangerous transformation. A long-standing left-to-right shunt exposes pulmonary vessels to excess flow and pressure. Pulmonary vascular remodeling raises pulmonary vascular resistance. When right-sided pressure exceeds left-sided pressure, flow reverses from right to left, causing cyanosis. The key management implication is that closure of the original defect may become harmful or contraindicated once severe irreversible pulmonary vascular disease has developed.

Atrial Septal Defect

Pathophysiology

An atrial septal defect is an opening in the interatrial septum that permits communication between the left atrium and right atrium. Because left atrial pressure is normally higher than right atrial pressure, oxygenated blood passes from the left atrium into the right atrium. The right atrium, right ventricle, and pulmonary arteries then receive extra volume. Over years, this chronic volume load can enlarge the right atrium and right ventricle, increase pulmonary blood flow, and predispose to atrial arrhythmias. The uploaded source notes that right-sided dilation becomes more likely as shunt size increases, particularly when the pulmonary-to-systemic flow ratio exceeds 1.5:1.

The most common ASD is the ostium secundum defect, which occurs in the central portion of the interatrial septum and accounts for about 80% of cases in the uploaded section. Ostium primum defects occur lower in the septum and are often linked with atrioventricular septal abnormalities. Sinus venosus defects occur high in the septum and may be associated with anomalous pulmonary venous drainage.

Clinical presentation

Many adults with ASD are asymptomatic for years because the pressure gradient between the atria is low and the shunt is often well tolerated. Symptoms often emerge in adulthood as right heart dilation, pulmonary overcirculation, or arrhythmias develop. The source highlights that patients are commonly asymptomatic until middle age, around 40 years old, and then develop exercise intolerance, dyspnea on exertion, or fatigue.

The classic exam finding is a fixed split S2. Normally, the split between aortic and pulmonic valve closure varies with respiration. In ASD, right-sided volume load delays pulmonic valve closure relatively constantly, so the split remains fixed. A mild systolic ejection murmur may be heard at the left upper sternal border or pulmonary area because of increased flow across the pulmonic valve. A diastolic flow rumble across the tricuspid valve may occur when right-sided flow is high. In advanced disease, signs of right-sided heart failure such as hepatomegaly, ascites, edema, or elevated JVP may appear.

Diagnosis and complications

Echocardiography is the central diagnostic test. Transthoracic echocardiography often identifies right-sided chamber enlargement and the defect, while transesophageal echocardiography can define septal anatomy more clearly. A bubble study can detect interatrial communication when agitated saline microbubbles appear on the left side after right-sided injection. Chest radiography may show large pulmonary arteries and increased pulmonary vascular markings. ECG may show right bundle branch block, right axis deviation, or atrial arrhythmias such as atrial fibrillation or flutter.

Complications of ASD include pulmonary hypertension, right heart failure, atrial arrhythmias, paradoxical embolism, and stroke. Pulmonary hypertension is generally a late complication in ASD rather than an early finding. Stroke can occur

through atrial fibrillation or paradoxical embolism if venous clot crosses the interatrial defect into systemic circulation.

Management

Not every ASD requires closure. A tiny asymptomatic defect without right-sided enlargement may be observed. Closure is considered when the defect is hemodynamically significant, particularly with symptoms, right atrial or right ventricular enlargement, or a Qp:Qs ratio above about 1.5:1 when pulmonary vascular resistance is acceptable. The uploaded source lists surgical repair when Qp:Qs is greater than 1.5:1 or 2:1 or when the patient is symptomatic. In modern practice, many secundum ASDs can be closed percutaneously when anatomy is favorable, while primum and sinus venosus defects more often require surgery.

Ventricular Septal Defect

Pathophysiology

A ventricular septal defect is an opening between the left and right ventricles. Because left ventricular pressure is normally much higher than right ventricular pressure, blood flows from left to right through the defect after birth. This sends oxygenated blood back through the pulmonary circulation, increasing pulmonary blood flow and loading the left atrium and left ventricle after the blood returns from the lungs. Large VSDs can also increase pulmonary artery pressures and eventually cause pulmonary vascular remodeling.

VSD is highlighted in the uploaded source as the most common congenital cardiac malformation. Small defects may close spontaneously and may create little hemodynamic burden. Large defects can produce congestive heart failure, growth failure in children, recurrent respiratory infections, pulmonary hypertension, and eventually Eisenmenger physiology if pulmonary vascular resistance rises enough to reverse the shunt.

Murmur and clinical clues

The classic VSD murmur is a harsh holosystolic murmur at the left lower sternal border, often around the fourth left intercostal space. A palpable thrill may be present. One of the counterintuitive exam associations is that a smaller restrictive VSD can be louder because the high-velocity jet across a small opening generates more turbulence. A very large nonrestrictive VSD may have a softer murmur because ventricular pressures become more equal. The uploaded source emphasizes that the smaller the defect, the louder the holosystolic murmur may be.

Handgrip can increase the murmur because increased systemic vascular resistance raises left-sided pressure and promotes more left-to-right flow across the VSD. Standing can reduce venous return and may reduce the murmur. As pulmonary vascular resistance increases, the pulmonic component of S2 may become louder. Aortic regurgitation can occur in some VSDs when the defect distorts aortic valve support.

Diagnosis, complications, and treatment

Echocardiography demonstrates the septal defect, estimates shunt size, evaluates chamber enlargement, and screens for pulmonary hypertension. Chest radiography may show cardiomegaly and increased pulmonary vascular markings in large left-to-right shunts. If pulmonary vascular resistance becomes high and the left-to-right shunt decreases, the heart size may appear less enlarged while the pulmonary arteries remain prominent.

Complications include infective endocarditis, progressive aortic regurgitation, heart failure, pulmonary hypertension, and Eisenmenger syndrome. Uncomplicated VSD without prior endocarditis does not automatically require endocarditis prophylaxis under modern prophylaxis rules. Repair is considered for hemodynamically significant shunts, symptoms, complications, or Qp:Qs thresholds such as greater than 1.5:1 or 2:1 when pulmonary vascular disease is not prohibitive. An asymptomatic patient with a small restrictive VSD often does not require surgery.

Coarctation of the Aorta

Pathophysiology and associations

Coarctation of the aorta is a narrowing of the aorta, classically near the origin of the left subclavian artery around the ductus arteriosus or ligamentum arteriosum. The narrowing creates obstruction between the proximal and distal aorta.

Proximal to the coarctation, pressure is high; distal to the coarctation, pressure is lower. The left ventricle faces increased afterload and may develop hypertrophy. Collateral vessels enlarge over time to bypass the obstruction.

The uploaded source emphasizes the adult exam pattern: hypertension in the upper extremities with hypotension in the lower extremities, delayed femoral pulses compared with radial pulses, and a midsystolic murmur heard best over the back. In women, coarctation may be associated with Turner syndrome. Coarctation is also associated with bicuspid aortic valve, although that association belongs in the broader congenital-aortic disease network as well.

Clinical presentation and diagnosis

Adults with coarctation may present with refractory hypertension, headache, exertional leg fatigue, cold lower extremities, claudication, or a murmur. The pulse exam is high yield: radial pulses arrive before femoral pulses, producing a radiofemoral delay. Blood pressure should be measured in both arms and at least one leg when coarctation is suspected.

CT angiography or MR angiography defines the anatomy. ECG may show left ventricular hypertrophy. Chest radiography can show inferior rib notching from enlarged intercostal collateral arteries and a figure-3 appearance from indentation at the narrowed segment with pre- and post-stenotic dilation.

Complications and treatment

Major complications include severe hypertension, premature coronary disease, aortic aneurysm, aortic dissection, rupture of cerebral aneurysms, infective endocarditis, and heart failure. Treatment may involve surgical repair, balloon angioplasty, or stenting depending on anatomy and patient characteristics. Adults require long-term surveillance because hypertension, recoarctation, aneurysm formation, and associated aortic valve disease can persist or appear after repair.

Patent Ductus Arteriosus

Physiology

The ductus arteriosus is a fetal vascular connection between the pulmonary artery and the aorta. In fetal life, it allows blood to bypass the nonfunctioning lungs. Prostaglandins and low oxygen tension help keep it open. After birth, rising oxygen tension and falling prostaglandin signaling normally lead to closure within days. When the ductus remains patent, postnatal systemic pressure in the aorta drives blood into the pulmonary artery, creating a left-to-right shunt.

A persistent PDA increases pulmonary blood flow and can produce left-sided volume overload because blood recirculates through the lungs and returns to the left heart. Over time, a large PDA can cause pulmonary hypertension, right-sided heart failure, cyanosis, and eventual shunt reversal. The uploaded source links PDA with congenital rubella syndrome, high altitude, and prematurity.

Clinical features

The classic PDA murmur is a continuous machinery murmur heard best at the left second intercostal space. Because blood flows continuously from the high-pressure aorta into the pulmonary artery during both systole and diastole, the murmur spans both phases. Patients may have wide pulse pressure and bounding peripheral pulses due to diastolic runoff into the pulmonary circulation. A loud P2 suggests pulmonary hypertension. Differential cyanosis can occur when severe pulmonary hypertension reverses flow so that deoxygenated blood enters the descending aorta distal to vessels supplying the upper body; toes may be more cyanotic than fingers.

Chest radiography can show increased pulmonary vascular markings, a dilated pulmonary artery, an enlarged cardiac silhouette, and sometimes calcification of the ductus. Echocardiography shows the patent ductus and turbulent flow.

Management

Management depends on pulmonary vascular disease. If pulmonary vascular disease is absent, closure by catheter-based or surgical techniques is considered for clinically significant PDA. The uploaded source uses surgical ligation as the classic treatment when pulmonary vascular disease is absent. If severe pulmonary hypertension or right-to-left shunting has developed, closure may be contraindicated because the ductus has become a pressure-release pathway for the right heart and pulmonary circulation.

The pharmacology is a classic anchor: indomethacin or other prostaglandin synthesis inhibitors can promote ductal closure in appropriate neonatal settings, while prostaglandin E1 can keep the ductus open when systemic or pulmonary blood flow depends on it, such as in certain ductal-dependent congenital heart lesions. In adult internal medicine, the key association is less about neonatal dosing and more about recognizing the physiology: PDA = aorta-to-pulmonary artery connection = continuous murmur = wide pulse pressure = left-to-right shunt unless pulmonary hypertension reverses it.

Tetralogy of Fallot

Anatomy and physiology

Tetralogy of Fallot consists of four abnormalities: ventricular septal defect, right ventricular hypertrophy, pulmonary stenosis or atresia, and overriding aorta. The common developmental problem involves abnormal formation of the outflow tract, especially the infundibular septum. The severity of symptoms depends largely on the degree of right ventricular outflow obstruction. If pulmonary stenosis is severe, blood preferentially crosses the VSD into the overriding aorta, creating right-to-left shunting and cyanosis.

The most common presenting symptom is cyanosis. Tet spells are episodic increases in cyanosis, often triggered by crying, exertion, feeding, or agitation. Squatting improves symptoms because it increases systemic vascular resistance. Higher SVR reduces right-to-left shunting across the VSD and encourages more blood to enter the pulmonary circulation. Oxygen, morphine, fluids, and beta-blockers may be used acutely in severe spells depending on the clinical setting, but definitive therapy is surgical.

Diagnosis and adult follow-up

Echocardiography is the diagnostic modality of choice because it defines the four anatomic abnormalities and provides information about right ventricular outflow tract obstruction and aortic anatomy. ECG may show right atrial and right ventricular enlargement. Chest radiography classically shows a boot-shaped heart due to right ventricular hypertrophy and an upturned apex. Cardiac catheterization or advanced imaging may be needed in selected patients to fully define anatomy.

Most patients undergo surgical repair in early life, and the uploaded source states that most patients have surgery within the first year and that 20-year survival after surgery is above 80%. Adult internal medicine therefore often sees repaired TOF rather than unrepaired TOF. Late issues include pulmonary regurgitation, residual right ventricular outflow obstruction, right ventricular dilation, atrial or ventricular arrhythmias, heart failure, and sudden cardiac death. This makes lifelong congenital cardiology follow-up important even when childhood repair was successful.

Original diagram: Shunt progression map

Left-to-right shunt progression

ASD, VSD, or PDA begins with oxygenated systemic-side blood recirculating to the pulmonary circulation. Increased pulmonary blood flow causes chamber dilation and pulmonary vascular stress. If the shunt is large and persistent, pulmonary vascular resistance can rise. When pulmonary vascular resistance exceeds systemic resistance, the shunt can reverse. Shunt reversal produces cyanosis and Eisenmenger physiology, at which point closure of the defect may be dangerous or contraindicated.

Comparison table: high-yield congenital lesions

Lesion	Anatomic problem	Classic clue	Major complications	Treatment logic
ASD	Interatrial communication; usually LA to RA flow	Fixed split S2; systolic flow murmur at pulmonary area; RA/RV dilation	AFib/flutter, right HF, pulmonary HTN, paradoxical embolus, Eisenmenger late	Observe small defects; close significant shunts if symptomatic or RA/RV enlarged and PVR acceptable
VSD	Interventricular communication; usually LV to RV flow	Harsh holosystolic murmur at LLSB; smaller can be louder	CHF, pulmonary HTN, Eisenmenger, endocarditis, aortic regurgitation	Observe small asymptomatic defects; repair significant shunts or complications

Lesion	Anatomic problem	Classic clue	Major complications	Treatment logic
Coarctation	Aortic narrowing near left subclavian/ductal region	Upper-extremity HTN, lower-extremity hypotension, delayed femoral pulses, rib notching	Severe HTN, cerebral aneurysm rupture, dissection, endocarditis	Surgery, angioplasty, or stenting; lifelong BP and aortic surveillance
PDA	Persistent aorta-to-pulmonary artery connection	Continuous machinery murmur, wide pulse pressure, bounding pulses	HF, pulmonary HTN, differential cyanosis, endocarditis	Close if significant and PVR acceptable; do not close severe irreversible pulmonary HTN with R-to-L shunt
TOF	VSD, RVH, pulmonary stenosis/atresia, overriding aorta	Cyanosis, Tet spells, squatting, boot-shaped heart	Arrhythmias, pulmonary regurgitation after repair, HF, sudden death	Surgical repair, usually early; lifelong follow-up after repair

Graph-RAG relationship map

For graph construction, congenital heart disease nodes should not be isolated facts. They should connect anatomy, hemodynamics, exam findings, diagnostic tests, complications, and management thresholds. For example, ASD should connect to left-to-right atrial shunt, RA/RV dilation, fixed split S2, atrial fibrillation, paradoxical embolus, Qp:Qs, and closure. VSD should connect to holosystolic murmur, pulmonary overcirculation, Eisenmenger syndrome, and endocarditis. Coarctation should connect to proximal hypertension, distal hypoperfusion, radiofemoral delay, rib notching, Turner syndrome, and aortic dissection. PDA should connect to prostaglandins, indomethacin, machinery murmur, bounding pulses, differential cyanosis, and endarteritis/endocarditis. TOF should connect to decreased pulmonary blood flow, right-to-left shunt, cyanosis, squatting, boot-shaped heart, and repaired-TOF late arrhythmias.

Illness scripts

Presentation	Most likely lesion	Why
Adult with dyspnea, fixed split S2, RBBB, right heart enlargement	ASD	Chronic left-to-right atrial shunt causes right-sided volume load
Child or adult with harsh holosystolic murmur at LLSB and thrill	VSD	Turbulent LV-to-RV systolic flow through septal defect
Young adult with hypertension in arms, weak femoral pulses, leg fatigue	Coarctation	Aortic narrowing obstructs distal flow and raises proximal pressure
Continuous murmur at left upper sternal border with bounding pulses	PDA	Aorta-to-pulmonary artery flow persists through systole and diastole
Cyanotic spells relieved by squatting; boot-shaped heart	Tetralogy of Fallot	Squatting raises SVR and reduces right-to-left shunt across VSD

Common pitfalls

A common pitfall is to think of all congenital heart disease as childhood-only. In reality, adults may present after decades of compensation, especially with ASD, coarctation, or repaired TOF complications. Another pitfall is to close every shunt automatically. Closure can be appropriate when a shunt is significant and pulmonary vascular resistance is acceptable, but closure may be contraindicated after severe irreversible pulmonary hypertension or right-to-left shunting has developed.

A third pitfall is confusing murmur intensity with lesion severity. In VSD, a smaller defect can create a louder murmur because the jet is high velocity, while a large defect can become softer if pressures equalize. A fourth pitfall is missing coarctation in a patient labeled with essential hypertension. Femoral pulses and arm-leg blood pressure comparison can reveal the diagnosis. A fifth pitfall is forgetting that adult repaired TOF is not cured in the sense of no follow-up; late pulmonary regurgitation, right ventricular dilation, arrhythmia, and sudden death risk require lifelong surveillance.

Practice questions

Question 1

A 42-year-old woman has progressive exertional dyspnea. Exam shows a fixed split S2 and a soft systolic ejection murmur at the left upper sternal border. ECG shows right bundle branch block. Echocardiography shows right atrial and right ventricular enlargement.

Answer: Atrial septal defect. **Teaching point:** ASD causes left-to-right atrial shunting, right-sided volume overload, fixed split S2, and adult-onset dyspnea or atrial arrhythmias.

Question 2

A patient has a harsh holosystolic murmur at the left lower sternal border with a palpable thrill. The murmur becomes louder with handgrip. The patient is otherwise asymptomatic.

Answer: Ventricular septal defect. **Teaching point:** VSD produces turbulent systolic flow from LV to RV. Smaller restrictive defects can be loud and may not require surgery if asymptomatic and uncomplicated.

Question 3

A 24-year-old woman has hypertension despite two medications. Arm blood pressure is elevated, leg blood pressure is lower, and femoral pulses are delayed. Chest x-ray shows inferior rib notching.

Answer: Coarctation of the aorta. **Teaching point:** Coarctation produces proximal hypertension, distal hypoperfusion, collateral intercostal enlargement, and radiofemoral delay.

Question 4

A patient has a continuous machinery murmur at the left second intercostal space, wide pulse pressure, and bounding pulses. Echocardiography shows flow from the aorta into the pulmonary artery.

Answer: Patent ductus arteriosus. **Teaching point:** PDA is a persistent aorta-pulmonary artery connection. It usually begins as a left-to-right shunt and can cause pulmonary overcirculation.

Question 5

A child with cyanotic congenital heart disease squats after exertion. Squatting improves symptoms. Imaging shows a VSD, RV hypertrophy, pulmonary stenosis, and overriding aorta.

Answer: Tetralogy of Fallot. **Teaching point:** Squatting raises systemic vascular resistance, reducing right-to-left shunting and increasing pulmonary blood flow.

Chapter summary

Adult congenital heart disease is best understood through hemodynamics. ASD, VSD, and PDA usually begin as left-to-right shunts that increase pulmonary blood flow and load right-sided or left-sided chambers depending on the lesion. Over time, large shunts can lead to pulmonary hypertension and Eisenmenger physiology, where shunt reversal creates cyanosis and changes management. Coarctation is an obstructive aortic lesion that causes upper-extremity hypertension, lower-extremity hypoperfusion, delayed femoral pulses, and collateral-related rib notching. Tetralogy of Fallot is a cyanotic lesion defined by VSD, RV hypertrophy, pulmonary stenosis or atresia, and overriding aorta; adults often present as repaired TOF patients with late arrhythmia, pulmonary regurgitation, or heart failure concerns. The highest-yield exam task is to link anatomy to murmur, chamber burden, pulmonary blood flow, cyanosis, diagnostic test, and closure or repair logic.

Source and reference notes

This chapter uses the user-uploaded congenital heart disease section as the coverage checklist and rewrites the material in original language for Medvale. The uploaded section covers ASD, VSD, coarctation of the aorta, PDA, and tetralogy of Fallot, including the numeric anchors used here. Modern adult congenital management framing was cross-checked against the 2018 AHA/ACC adult congenital heart disease guideline and the 2020 ESC adult congenital heart disease guideline. This chapter is educational and is not a substitute for clinical care or local specialist guidance.